

ANALYTICAL REPORT

PREPARED FOR

Attn: Cindy Tomaszewski
City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Generated 5/17/2023 1:35:30 PM Revision 2

JOB DESCRIPTION

City of Cadillac BioSolids PFAS 2023

JOB NUMBER

190-31485-1

Eurofins Michigan

Job Notes

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The test results in this report relate only to the samples as received by the laboratory and will meet all requirements of the methodology, with any exceptions noted. This report shall not be reproduced except in full, without the express written approval of the laboratory. All questions should be directed to the Eurofins Environment Testing North Central, LLC Project Manager.

Authorization



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Sample Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-31485-1	BioSolids	Solid	04/13/23 14:15	04/14/23 10:48
190-31485-2	EQUIPMENT BLANK	Water	04/13/23 14:15	04/14/23 10:48

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12

Case Narrative

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Job ID: 190-31485-1

Laboratory: Eurofins Michigan

Narrative

Job Narrative
190-31485-1

REVISION

The report being provided is a revision of the original report sent on 5/12/2023 which has been recalculated to the MDL.

Receipt

The samples were received on 4/14/2023 10:48 AM. Unless otherwise noted below, the samples arrived in good condition, and, where required, properly preserved and on ice. The temperature of the cooler at receipt time was 4.0°C

PFAS

Method PFC_IDA: The laboratory control sample (LCS) for preparation batch 320-669632 and analytical batch 320-670627 recovered outside control limits for the following analyte: 11CI-PF3OUdS. This analyte was biased high in the LCS and were not detected in the associated samples; therefore, the data have been reported: EQUIPMENT BLANK (190-31485-2), (LCS 320-669632/2-A) and (240-183540-M-3-A).

Method PFC_IDA: The following samples in preparation batch 320-670155 were yellow in color following extraction. BioSolids (190-31485-1) preparation batch 320-670155 Method: PFC_IDAMatrix: Solids

No additional analytical or quality issues were noted, other than those described above or in the Definitions/ Glossary page.

General Chemistry

No additional analytical or quality issues were noted, other than those described above or in the Definitions/ Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Client Sample ID: BioSolids

Lab Sample ID: 190-31485-1

Date Collected: 04/13/23 14:15

Matrix: Solid

Date Received: 04/14/23 10:48

Percent Solids: 2.6

Method: EPA 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
F-53B Major	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
F-53B Minor	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
4:2 FTS	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
6:2 FTS	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
8:2 FTS	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
HFPO-DA (GenX)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorobutanesulfonic acid (PFBS)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorobutanoic acid (PFBA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorodecanesulfonic acid (PFDS)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorodecanoic acid (PFDA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorododecanoic acid (PFDoA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluoroheptanesulfonic acid (PFHpS)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluoroheptanoic acid (PFHpA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorohexanesulfonic acid (PFHxS)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorohexanoic acid (PFHxA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorononanesulfonic acid (PFNS)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorononanoic acid (PFNA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorooctanesulfonamide (FOSA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorooctanesulfonic acid (PFOS)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorooctanoic acid (PFOA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluoropentanesulfonic acid (PFPeS)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluoropentanoic acid (PFPeA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorotetradecanoic acid (PFTeA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluorotridecanoic acid (PFTriA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1
Perfluoroundecanoic acid (PFUnA)	<7.5		7.5	ug/Kg	☆	04/25/23 20:30	04/29/23 00:16	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	107		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C3 HFPO-DA	92		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C4 PFBA	93		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C3 PFBS	88		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C2 PFDA	119		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C2 PFDoA	109		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C4 PFHpA	107		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C2 PFHxA	108		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C5 PFNA	106		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C4 PFOA	102		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C4 PFOS	90		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C5 PFPeA	99		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C2 PFTeA	69		25 - 150	04/25/23 20:30	04/29/23 00:16	1
13C2 PFUnA	122		25 - 150	04/25/23 20:30	04/29/23 00:16	1
d5-NEtFOSAA	123		25 - 150	04/25/23 20:30	04/29/23 00:16	1
d3-NMeFOSAA	114		25 - 150	04/25/23 20:30	04/29/23 00:16	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Client Sample ID: BioSolids

Lab Sample ID: 190-31485-1

Date Collected: 04/13/23 14:15

Matrix: Solid

Date Received: 04/14/23 10:48

Percent Solids: 2.6

Method: EPA 537 (modified) - Fluorinated Alkyl Substances (Continued)

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
M2-4:2 FTS	135		25 - 150	04/25/23 20:30	04/29/23 00:16	1
M2-6:2 FTS	130		25 - 150	04/25/23 20:30	04/29/23 00:16	1
M2-8:2 FTS	142		25 - 150	04/25/23 20:30	04/29/23 00:16	1
18O2 PFHxS	88		25 - 150	04/25/23 20:30	04/29/23 00:16	1

General Chemistry

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Percent Moisture (ASTM D 2216)	97.4	H	0.1	0.1 %			05/15/23 14:40	1
Percent Solids (ASTM D 2216)	2.6	H	0.1	0.1 %			05/15/23 14:40	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Client Sample ID: EQUIPMENT BLANK

Lab Sample ID: 190-31485-2

Date Collected: 04/13/23 14:15

Matrix: Water

Date Received: 04/14/23 10:48

Method: EPA 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
F-53B Major	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
F-53B Minor	<1.9	+	1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
4:2 FTS	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
6:2 FTS	<4.6		4.6	ng/L		04/24/23 05:50	04/27/23 14:36	1
8:2 FTS	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
HFPO-DA (GenX)	<3.7		3.7	ng/L		04/24/23 05:50	04/27/23 14:36	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<4.6		4.6	ng/L		04/24/23 05:50	04/27/23 14:36	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<4.6		4.6	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorobutanesulfonic acid (PFBS)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorobutanoic acid (PFBA)	<4.6		4.6	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorodecanesulfonic acid (PFDS)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorodecanoic acid (PFDA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorododecanoic acid (PFDoA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluoroheptanesulfonic acid (PFHpS)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluoroheptanoic acid (PFHpA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorohexanesulfonic acid (PFHxS)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorohexanoic acid (PFHxA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorononanesulfonic acid (PFNS)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorononanoic acid (PFNA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorooctanesulfonamide (FOSA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorooctanesulfonic acid (PFOS)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorooctanoic acid (PFOA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluoropentanesulfonic acid (PFPeS)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluoropentanoic acid (PFPeA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorotetradecanoic acid (PFTeA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluorotridecanoic acid (PFTriA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1
Perfluoroundecanoic acid (PFUnA)	<1.9		1.9	ng/L		04/24/23 05:50	04/27/23 14:36	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	94		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C3 HFPO-DA	113		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C4 PFBA	102		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C3 PFBS	106		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C2 PFDA	100		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C2 PFDoA	99		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C4 PFHpA	108		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C2 PFHxA	96		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C5 PFNA	102		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C4 PFOA	104		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C4 PFOS	100		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C5 PFPeA	120		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C2 PFTeA	91		25 - 150	04/24/23 05:50	04/27/23 14:36	1
13C2 PFUnA	102		25 - 150	04/24/23 05:50	04/27/23 14:36	1
d5-NEtFOSAA	106		25 - 150	04/24/23 05:50	04/27/23 14:36	1
d3-NMeFOSAA	99		25 - 150	04/24/23 05:50	04/27/23 14:36	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Client Sample ID: EQUIPMENT BLANK

Lab Sample ID: 190-31485-2

Date Collected: 04/13/23 14:15

Matrix: Water

Date Received: 04/14/23 10:48

Method: EPA 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
M2-4:2 FTS	106		25 - 150	04/24/23 05:50	04/27/23 14:36	1
M2-6:2 FTS	86		25 - 150	04/24/23 05:50	04/27/23 14:36	1
M2-8:2 FTS	95		25 - 150	04/24/23 05:50	04/27/23 14:36	1
18O2 PFHxS	90		25 - 150	04/24/23 05:50	04/27/23 14:36	1

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 320-669632/1-A

Matrix: Water

Analysis Batch: 670627

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 669632

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
F-53B Major	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
F-53B Minor	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
4:2 FTS	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
6:2 FTS	<5.0		5.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
8:2 FTS	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
HFPO-DA (GenX)	<4.0		4.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<5.0		5.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<5.0		5.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorobutanoic acid (PFBA)	<5.0		5.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluoroheptanesulfonic acid (PFHpS)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorononanoic acid (PFNA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorooctanesulfonamide (FOSA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorooctanesulfonic acid (PFOS)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluorotridecanoic acid (PFTriA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0	ng/L		04/24/23 05:50	04/27/23 13:55	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	95		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C3 HFPO-DA	106		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C4 PFBA	102		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C3 PFBS	111		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C2 PFDA	98		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C2 PFDoA	109		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C4 PFHpA	115		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C2 PFHxA	101		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C5 PFNA	99		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C4 PFOA	100		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C4 PFOS	97		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C5 PFPeA	112		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C2 PFTeA	96		25 - 150	04/24/23 05:50	04/27/23 13:55	1
13C2 PFUnA	100		25 - 150	04/24/23 05:50	04/27/23 13:55	1
d5-NEtFOSAA	109		25 - 150	04/24/23 05:50	04/27/23 13:55	1

Eurofins Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 320-669632/1-A

Matrix: Water

Analysis Batch: 670627

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 669632

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
d3-NMeFOSAA	98		25 - 150	04/24/23 05:50	04/27/23 13:55	1
M2-4:2 FTS	104		25 - 150	04/24/23 05:50	04/27/23 13:55	1
M2-6:2 FTS	82		25 - 150	04/24/23 05:50	04/27/23 13:55	1
M2-8:2 FTS	84		25 - 150	04/24/23 05:50	04/27/23 13:55	1
18O2 PFHxS	101		25 - 150	04/24/23 05:50	04/27/23 13:55	1

Lab Sample ID: LCS 320-669632/2-A

Matrix: Water

Analysis Batch: 670627

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 669632

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	37.8	50.4		ng/L		134	79 - 139
F-53B Major	37.4	43.7		ng/L		117	75 - 135
F-53B Minor	37.8	45.3	*+	ng/L		120	54 - 114
4:2 FTS	37.5	44.1		ng/L		118	79 - 139
6:2 FTS	38.1	36.9		ng/L		97	59 - 175
8:2 FTS	38.4	38.9		ng/L		101	75 - 135
HFPO-DA (GenX)	40.0	42.4		ng/L		106	51 - 173
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	41.9		ng/L		105	76 - 136
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	43.0		ng/L		107	76 - 136
Perfluorobutanesulfonic acid (PFBS)	35.5	39.7		ng/L		112	67 - 127
Perfluorobutanoic acid (PFBA)	40.0	42.3		ng/L		106	76 - 136
Perfluorodecanesulfonic acid (PFDS)	38.6	44.1		ng/L		114	71 - 131
Perfluorodecanoic acid (PFDA)	40.0	48.0		ng/L		120	76 - 136
Perfluorododecanoic acid (PFDoA)	40.0	40.7		ng/L		102	71 - 131
Perfluoroheptanesulfonic acid (PFHpS)	38.2	46.3		ng/L		121	76 - 136
Perfluoroheptanoic acid (PFHpA)	40.0	38.7		ng/L		97	72 - 132
Perfluorohexanesulfonic acid (PFHxS)	36.5	38.7		ng/L		106	59 - 119
Perfluorohexanoic acid (PFHxA)	40.0	39.1		ng/L		98	73 - 133
Perfluorononanesulfonic acid (PFNS)	38.5	46.3		ng/L		120	75 - 135
Perfluorononanoic acid (PFNA)	40.0	49.2		ng/L		123	75 - 135
Perfluorooctanesulfonamide (FOSA)	40.0	43.1		ng/L		108	73 - 133
Perfluorooctanesulfonic acid (PFOS)	37.2	39.3		ng/L		106	70 - 130
Perfluorooctanoic acid (PFOA)	40.0	42.4		ng/L		106	70 - 130
Perfluoropentanesulfonic acid (PFPeS)	37.6	40.2		ng/L		107	66 - 126
Perfluoropentanoic acid (PFPeA)	40.0	46.3		ng/L		116	71 - 131
Perfluorotetradecanoic acid (PFTeA)	40.0	39.3		ng/L		98	70 - 130
Perfluorotridecanoic acid (PFTriA)	40.0	40.1		ng/L		100	71 - 131

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-669632/2-A

Matrix: Water

Analysis Batch: 670627

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 669632

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec Limits
Perfluoroundecanoic acid (PFUnA)	40.0	45.5		ng/L		114	68 - 128
Isotope Dilution	LCS %Recovery	LCS Qualifier	Limits				
13C8 FOSA	106		25 - 150				
13C3 HFPO-DA	114		25 - 150				
13C4 PFBA	114		25 - 150				
13C3 PFBS	115		25 - 150				
13C2 PFDA	109		25 - 150				
13C2 PFDoA	112		25 - 150				
13C4 PFHpA	118		25 - 150				
13C2 PFHxA	116		25 - 150				
13C5 PFNA	102		25 - 150				
13C4 PFOA	111		25 - 150				
13C4 PFOS	100		25 - 150				
13C5 PFPeA	124		25 - 150				
13C2 PFTeDA	105		25 - 150				
13C2 PFUnA	110		25 - 150				
d5-NEtFOSAA	112		25 - 150				
d3-NMeFOSAA	118		25 - 150				
M2-4:2 FTS	101		25 - 150				
M2-6:2 FTS	98		25 - 150				
M2-8:2 FTS	102		25 - 150				
18O2 PFHxS	107		25 - 150				

Lab Sample ID: MB 320-670155/1-A

Matrix: Solid

Analysis Batch: 670858

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 670155

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
F-53B Major	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
F-53B Minor	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
4:2 FTS	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
6:2 FTS	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
8:2 FTS	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
HFPO-DA (GenX)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorobutanesulfonic acid (PFBS)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorobutanoic acid (PFBA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorodecanesulfonic acid (PFDS)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorodecanoic acid (PFDA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorododecanoic acid (PFDoA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluoroheptanesulfonic acid (PFHpS)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluoroheptanoic acid (PFHpA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 320-670155/1-A

Matrix: Solid

Analysis Batch: 670858

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 670155

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Perfluorohexanesulfonic acid (PFHxS)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorohexanoic acid (PFHxA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorononanesulfonic acid (PFNS)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorononanoic acid (PFNA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorooctanesulfonamide (FOSA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorooctanesulfonic acid (PFOS)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorooctanoic acid (PFOA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluoropentanesulfonic acid (PFPeS)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluoropentanoic acid (PFPeA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorotetradecanoic acid (PFTeA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluorotridecanoic acid (PFTriA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1
Perfluoroundecanoic acid (PFUnA)	<0.20		0.20	ug/Kg		04/25/23 20:30	04/28/23 23:56	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	86		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C3 HFPO-DA	85		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C4 PFBA	84		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C3 PFBS	79		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C2 PFDA	99		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C2 PFDoA	99		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C4 PFHpA	93		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C2 PFHxA	91		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C5 PFNA	92		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C4 PFOA	90		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C4 PFOS	82		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C5 PFPeA	86		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C2 PFTeDA	107		25 - 150	04/25/23 20:30	04/28/23 23:56	1
13C2 PFUnA	100		25 - 150	04/25/23 20:30	04/28/23 23:56	1
d5-NEtFOSAA	103		25 - 150	04/25/23 20:30	04/28/23 23:56	1
d3-NMeFOSAA	96		25 - 150	04/25/23 20:30	04/28/23 23:56	1
M2-4:2 FTS	90		25 - 150	04/25/23 20:30	04/28/23 23:56	1
M2-6:2 FTS	86		25 - 150	04/25/23 20:30	04/28/23 23:56	1
M2-8:2 FTS	111		25 - 150	04/25/23 20:30	04/28/23 23:56	1
18O2 PFHxS	81		25 - 150	04/25/23 20:30	04/28/23 23:56	1

Lab Sample ID: LCS 320-670155/2-A

Matrix: Solid

Analysis Batch: 670858

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 670155

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	1.89	2.53		ug/Kg		134	79 - 139
F-53B Major	1.87	2.29		ug/Kg		123	74 - 134
F-53B Minor	1.89	2.51		ug/Kg		133	66 - 136
4:2 FTS	1.88	2.17		ug/Kg		115	68 - 143
6:2 FTS	1.90	2.37		ug/Kg		124	73 - 139
8:2 FTS	1.92	2.16		ug/Kg		113	75 - 135
HFPO-DA (GenX)	2.00	2.24		ug/Kg		112	53 - 158

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-670155/2-A

Matrix: Solid

Analysis Batch: 670858

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 670155

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec Limits
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	2.00	1.97		ug/Kg		99	72 - 132
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	2.00	2.21		ug/Kg		111	72 - 132
Perfluorobutanesulfonic acid (PFBS)	1.78	1.88		ug/Kg		106	69 - 129
Perfluorobutanoic acid (PFBA)	2.00	2.14		ug/Kg		107	76 - 136
Perfluorodecanesulfonic acid (PFDS)	1.93	2.23		ug/Kg		116	71 - 131
Perfluorodecanoic acid (PFDA)	2.00	2.26		ug/Kg		113	72 - 132
Perfluorododecanoic acid (PFDoA)	2.00	2.20		ug/Kg		110	71 - 131
Perfluoroheptanesulfonic acid (PFHpS)	1.91	2.13		ug/Kg		112	76 - 136
Perfluoroheptanoic acid (PFHpA)	2.00	2.14		ug/Kg		107	71 - 131
Perfluorohexanesulfonic acid (PFHxS)	1.82	2.02		ug/Kg		111	62 - 122
Perfluorohexanoic acid (PFHxA)	2.00	1.98		ug/Kg		99	71 - 131
Perfluorononanesulfonic acid (PFNS)	1.92	2.12		ug/Kg		110	72 - 132
Perfluorononanoic acid (PFNA)	2.00	2.24		ug/Kg		112	73 - 133
Perfluorooctanesulfonamide (FOSA)	2.00	2.27		ug/Kg		113	77 - 137
Perfluorooctanesulfonic acid (PFOS)	1.86	1.95		ug/Kg		105	68 - 141
Perfluorooctanoic acid (PFOA)	2.00	2.20		ug/Kg		110	72 - 132
Perfluoropentanesulfonic acid (PFPeS)	1.88	2.20		ug/Kg		117	66 - 126
Perfluoropentanoic acid (PFPeA)	2.00	2.23		ug/Kg		112	69 - 129
Perfluorotetradecanoic acid (PFTeA)	2.00	1.79		ug/Kg		90	67 - 127
Perfluorotridecanoic acid (PFTriA)	2.00	2.16		ug/Kg		108	71 - 131
Perfluoroundecanoic acid (PFUnA)	2.00	2.37		ug/Kg		118	66 - 126

Isotope Dilution	LCS LCS		Limits
	%Recovery	Qualifier	
13C8 FOSA	87		25 - 150
13C3 HFPO-DA	83		25 - 150
13C4 PFBA	89		25 - 150
13C3 PFBS	81		25 - 150
13C2 PFDA	99		25 - 150
13C2 PFDoA	100		25 - 150
13C4 PFHpA	96		25 - 150
13C2 PFHxA	88		25 - 150
13C5 PFNA	92		25 - 150
13C4 PFOA	90		25 - 150
13C4 PFOS	81		25 - 150
13C5 PFPeA	87		25 - 150
13C2 PFTeDA	107		25 - 150
13C2 PFUnA	98		25 - 150
d5-NEtFOSAA	99		25 - 150

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QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-670155/2-A

Matrix: Solid

Analysis Batch: 670858

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 670155

Isotope Dilution	LCS		Limits
	%Recovery	Qualifier	
d3-NMeFOSAA	90		25 - 150
M2-4:2 FTS	95		25 - 150
M2-6:2 FTS	92		25 - 150
M2-8:2 FTS	110		25 - 150
18O2 PFHxS	83		25 - 150

Method: D 2216 - Percent Moisture

Lab Sample ID: 190-31485-1 DU

Matrix: Solid

Analysis Batch: 674867

Client Sample ID: BioSolids

Prep Type: Total/NA

Analyte	Sample		DU		Unit	D	RPD	RPD Limit
	Result	Qualifier	Result	Qualifier				
Percent Moisture	97.4	H	97.7		%		0.3	20
Percent Solids	2.6	H	2.3		%		13	20

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Solid

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (25-150)	PFDA (25-150)	PFDoA (25-150)	C4PFHA (25-150)	PFHxA (25-150)
190-31485-1	BioSolids	107	92	93	88	119	109	107	108
LCS 320-670155/2-A	Lab Control Sample	87	83	89	81	99	100	96	88
MB 320-670155/1-A	Method Blank	86	85	84	79	99	99	93	91

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (25-150)	PFOA (25-150)	PFOS (25-150)	PFPeA (25-150)	PFTDA (25-150)	PFUnA (25-150)	d5NEFOS (25-150)	d3NMFOS (25-150)
190-31485-1	BioSolids	106	102	90	99	69	122	123	114
LCS 320-670155/2-A	Lab Control Sample	92	90	81	87	107	98	99	90
MB 320-670155/1-A	Method Blank	92	90	82	86	107	100	103	96

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (25-150)
190-31485-1	BioSolids	135	130	142	88
LCS 320-670155/2-A	Lab Control Sample	95	92	110	83
MB 320-670155/1-A	Method Blank	90	86	111	81

Surrogate Legend

PFOSA = 13C8 FOSA
HFPODA = 13C3 HFPO-DA
PFBA = 13C4 PFBA
C3PFBS = 13C3 PFBS
PFDA = 13C2 PFDA
PFDoA = 13C2 PFDoA
C4PFHA = 13C4 PFHpA
PFHxA = 13C2 PFHxA
PFNA = 13C5 PFNA
PFOA = 13C4 PFOA
PFOS = 13C4 PFOS
PFPeA = 13C5 PFPeA
PFTDA = 13C2 PFTeDA
PFUnA = 13C2 PFUnA
d5NEFOS = d5-NEtFOSAA
d3NMFOS = d3-NMeFOSAA
M242FTS = M2-4:2 FTS
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS
PFHxS = 18O2 PFHxS

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (25-150)	PFDA (25-150)	PFDoA (25-150)	C4PFHA (25-150)	PFHxA (25-150)
190-31485-2	EQUIPMENT BLANK	94	113	102	106	100	99	108	96
LCS 320-669632/2-A	Lab Control Sample	106	114	114	115	109	112	118	116
MB 320-669632/1-A	Method Blank	95	106	102	111	98	109	115	101

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Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (25-150)	PFOA (25-150)	PFOS (25-150)	PFPeA (25-150)	PFTDA (25-150)	PFUnA (25-150)	d5NEFOS (25-150)	d3NMFOS (25-150)
190-31485-2	EQUIPMENT BLANK	102	104	100	120	91	102	106	99
LCS 320-669632/2-A	Lab Control Sample	102	111	100	124	105	110	112	118
MB 320-669632/1-A	Method Blank	99	100	97	112	96	100	109	98

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (25-150)
190-31485-2	EQUIPMENT BLANK	106	86	95	90
LCS 320-669632/2-A	Lab Control Sample	101	98	102	107
MB 320-669632/1-A	Method Blank	104	82	84	101

Surrogate Legend

PFOSA = 13C8 FOSA
HFPODA = 13C3 HFPO-DA
PFBA = 13C4 PFBA
C3PFBS = 13C3 PFBS
PFDA = 13C2 PFDA
PFDoA = 13C2 PFDoA
C4PFHA = 13C4 PFHpA
PFHxA = 13C2 PFHxA
PFNA = 13C5 PFNA
PFOA = 13C4 PFOA
PFOS = 13C4 PFOS
PFPeA = 13C5 PFPeA
PFTDA = 13C2 PFTeDA
PFUnA = 13C2 PFUnA
d5NEFOS = d5-NEtFOSAA
d3NMFOS = d3-NMeFOSAA
M242FTS = M2-4:2 FTS
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS
PFHxS = 18O2 PFHxS

Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Qualifiers

LCMS

Qualifier	Qualifier Description
*+	LCS and/or LCSD is outside acceptance limits, high biased.

General Chemistry

Qualifier	Qualifier Description
H	Sample was prepped or analyzed beyond the specified holding time. This does not meet regulatory requirements.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
□	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CFU	Colony Forming Unit
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MCL	EPA recommended "Maximum Contaminant Level"
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
MPN	Most Probable Number
MQL	Method Quantitation Limit
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
NEG	Negative / Absent
POS	Positive / Present
PQL	Practical Quantitation Limit
PRES	Presumptive
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)
TNTC	Too Numerous To Count

QC Association Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

LCMS

Prep Batch: 669632

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-31485-2	EQUIPMENT BLANK	Total/NA	Water	3535	
MB 320-669632/1-A	Method Blank	Total/NA	Water	3535	
LCS 320-669632/2-A	Lab Control Sample	Total/NA	Water	3535	

Prep Batch: 670155

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-31485-1	BioSolids	Total/NA	Solid	SHAKE	
MB 320-670155/1-A	Method Blank	Total/NA	Solid	SHAKE	
LCS 320-670155/2-A	Lab Control Sample	Total/NA	Solid	SHAKE	

Analysis Batch: 670627

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-31485-2	EQUIPMENT BLANK	Total/NA	Water	537 (modified)	669632
MB 320-669632/1-A	Method Blank	Total/NA	Water	537 (modified)	669632
LCS 320-669632/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	669632

Analysis Batch: 670858

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-31485-1	BioSolids	Total/NA	Solid	537 (modified)	670155
MB 320-670155/1-A	Method Blank	Total/NA	Solid	537 (modified)	670155
LCS 320-670155/2-A	Lab Control Sample	Total/NA	Solid	537 (modified)	670155

General Chemistry

Analysis Batch: 674867

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-31485-1	BioSolids	Total/NA	Solid	D 2216	
190-31485-1 DU	BioSolids	Total/NA	Solid	D 2216	

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Client Sample ID: BioSolids

Date Collected: 04/13/23 14:15

Date Received: 04/14/23 10:48

Lab Sample ID: 190-31485-1

Matrix: Solid

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Analyst	Lab	Prepared or Analyzed
Total/NA	Analysis	D 2216		1	674867	TCS	EET SAC	05/15/23 14:40

Client Sample ID: BioSolids

Date Collected: 04/13/23 14:15

Date Received: 04/14/23 10:48

Lab Sample ID: 190-31485-1

Matrix: Solid

Percent Solids: 2.6

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Analyst	Lab	Prepared or Analyzed
Total/NA	Prep	SHAKE			670155	FX	EET SAC	04/25/23 20:30
Total/NA	Analysis	537 (modified)		1	670858	D1R	EET SAC	04/29/23 00:16

Client Sample ID: EQUIPMENT BLANK

Date Collected: 04/13/23 14:15

Date Received: 04/14/23 10:48

Lab Sample ID: 190-31485-2

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Analyst	Lab	Prepared or Analyzed
Total/NA	Prep	3535			669632	EJR	EET SAC	04/24/23 05:50
Total/NA	Analysis	537 (modified)		1	670627	RS1	EET SAC	04/27/23 14:36

Laboratory References:

EET SAC = Eurofins Sacramento, 880 Riverside Parkway, West Sacramento, CA 95605, TEL (916)373-5600

Analyst References:

Lab: EET SAC

Batch Type: Prep

EJR = Eduardo Rangel

FX = Fong Xiong

Batch Type: Analysis

D1R = Dhatpakorn Ruangyotsakul

RS1 = Rungtip Sanjumnai

TCS = Tammy Saechao

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac BioSolids PFAS 2023

Job ID: 190-31485-1

Laboratory: Eurofins Sacramento

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	Identification Number	Expiration Date
Alaska (UST)	State	17-020	02-20-24
ANAB	Dept. of Defense ELAP	L2468	01-20-24
ANAB	Dept. of Energy	L2468.01	01-20-24
ANAB	ISO/IEC 17025	L2468	01-20-24
Arizona	State	AZ0708	08-11-23
Arkansas DEQ	State	88-0691	06-17-23
California	State	2897	01-22-24
Colorado	State	CA0004	08-31-23
Florida	NELAP	E87570	06-30-23
Georgia	State	4040	01-29-24
Hawaii	State	<cert No.>	01-29-24
Illinois	NELAP	200060	03-17-24
Kansas	NELAP	E-10375	10-31-23
Louisiana	NELAP	01944	06-30-23
Louisiana (All)	NELAP	01944	06-30-23
Maine	State	CA00004	04-14-24
Michigan	State	9947	06-01-23
Nevada	State	CA00044	07-31-23
New Hampshire	NELAP	2997	04-18-24
New Jersey	NELAP	CA005	06-30-23
New York	NELAP	11666	04-01-24
Ohio	State	41252	01-29-24
Oregon	NELAP	4040	01-29-24
Texas	NELAP	T104704399-19-13	05-31-23
US Fish & Wildlife	US Federal Programs	58448	04-30-24
USDA	US Federal Programs	P330-18-00239	02-28-26
Utah	NELAP	CA000442021-12	02-28-23 *
Virginia	NELAP	460278	03-14-24
Washington	State	C581	05-05-23 *
West Virginia (DW)	State	9930C	12-31-23
Wisconsin	State	998204680	08-31-23
Wyoming	State Program	8TMS-L	01-28-19 *

* Accreditation/Certification renewal pending - accreditation/certification considered valid.

Eurofins Michigan

Client Information		Lab PM: Schafer, Sue		Carrier Tracking No(s): 190-38324-2796, 1	
Client Contact: Cindy Tomaszewski		E-Mail: Sue.Schafer@et.eurofinus.com		Page: Page 1 of 1	
Company: City of Cadillac Utilities		PWSID: 231-775-2308		Job #:	
Address: 11121 Plett Road		Due Date Requested:		Analysis Requested	
City: Cadillac		TAT Requested (days):		Total Number of Containers	
State, Zip: MI, 49601		Compliance Project: ☒ Yes ☐ No		Preservation Codes:	
Phone:		PO #: 2019-045-047230413 TA		M - Hexane	
Email: lab@cadillac-mi.net		WO #: 19001884		N - None	
Project Name: City of Cadillac Biosolids PFAS 2023		Project #: 19001884		O - AsNaO2	
Site:		SSOW#:		P - Na2O4S	
Sample Identification		Sample Date		Q - Na2SO3	
Biosolids		4-13-23		R - Na2SO4	
EQUIPMENT BLANK		Sample Time		S - H2SO4	
		1415		T - TSP Dodecahydrate	
		Sample Type (C=Comp, G=grab)		U - Acetone	
		G		V - MCAA	
		Matrix (W=water, B=solid, O=wastewat, BT=tissue, A=air)		W - pH 4-5	
		Solid		Y - Trizma	
		Water		Z - other (specify)	
		Preservation Code:		Other:	
		Field Filtered Sample (Yes or No)		Special Instructions/Note:	
		Perform MS/MSD (Yes or No)			
		PFC, IJA - (MOD) PFAS 28			
		Moisture - Local Method			
		PFC, IJA - PFAS 28			
		Field Filtered Sample (Yes or No)			
		Perform MS/MSD (Yes or No)			
		PFC, IJA - (MOD) PFAS 28			
		Moisture - Local Method			
		PFC, IJA - PFAS 28			
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		PFC, IJA - PFAS 28			
		Field Filtered Sample (Yes or No)			
		Perform MS/MSD (Yes or No)			
		PFC, IJA - (MOD) PFAS 28			
		Moisture - Local Method			



Environment Testing
TestAmerica

- ☐ SDS or Known Hazard Information Supplied by Client
☐ Discrepancies
☐ Short Hold
☐ Rush ☐ 24 Hr ☐ 2-Day ☐ 3-Day ☐ 5-Day ☐ Other:

Client ID: City of Cadillac
Work Order #: 31485

Cooler / Sample Receipt

After hours receipt: complete gray areas. Place cooler in walk-in, place form in Receiving box. Date: _____ Time: _____

Receipt Evaluation Performed by: Initials: ML Date: 4/14 Time: 10:48

Method of Shipment:

Walk-In Client Eurofins TA Field/Courier

Other Client / 3rd Party Courier: _____

Fed Ex Tracking #: _____

UPS Tracking #: Ground

Other: _____

Shipping Container Type:

- ☒ Cooler ☒ Box
☐ None ☐ Other: _____

Packing Materials:

- ☐ Plastic Bags ☐ Foam
☐ Bubble Wrap ☐ Paper
☐ Packing Peanuts ☐ None
☐ Other: _____

Custody Seals Intact:

- ☒ Yes ☐ No
☐ NA (not used or required)

Cooling Materials:

- ☐ Ice (Solid) ☒ Ice (Melted)
☒ Blue Ice ☐ None
☐ Other: _____

Bacteriological Samples	Temp Corrected (°C)	Frozen?		Rec'd Within 2 Hrs?		Sample Flagged?	
		Yes	No	Yes	No	Yes	No

Received on same day sampled? Yes ☐ No ☒

Additional Sheets Required? Yes ☐ No ☒

Receipt Temperatures

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Acceptable	Cooler ID	Affected Samples
CP313207	4.0	4.0			Y N		
					Y N		
					Y N		

Receipt Questions**	Y	N	NA	"No" answers require additional comment
CoC present and ETA receipt signature, date, and time properly documented?		☒		No Sample time/date on 2nd Sample. Noted from label on the bag.
Containers and Labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)		☒		No labels on bottles. Labels only on plastic bags that come with.
Appropriate containers used and adequate volume provided?	☒			Preserved bottles checked for pH? Yes ☒ No ☐
Number of sample containers match CoC?	☒			pH strip lot # _____
Samples received within hold?	☒			
Samples submitted for GRO and Volatiles analysis (8260, 624, 524) received without headspace?			☒	
Was a Trip Blank received with VOA samples?			☒	
Were the samples free of any questionable physical conformities? (i.e.: field duplicates or multiple bottles of the same sample do not significantly vary in appearance - color, solid proportions, etc.)	☒			
Were the CoC bottle labels and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	☒			
**May not be applicable if samples are not for compliance testing				*Excludes FOG, VOAs, TOC Vials, HEM

Client Contact Record

Contact Via: ☐ Phone ☐ Email ☐ Other: _____ Person Contacted: _____ Date/Time: _____

☐ Discrepancy allowance agreement is on record in the client project file

Discussion / Resolution

Any additional documentation and clarification from the client must be noted in the narrative and/or scanned into the CoC directory.

Reviewed by ML Date: 4/14/2023

WI-MI-010_020720

